

Assignment 1

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March 13, 2026

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1 Task 1: Monte Carlo Simulation

1.1 The model

In this task I simulated the group stage for the 2022 FIFA World Cup using the given Competing Poisson-distribution model. The Poisson distribution is used to model the number of times an event occurs in a given interval, assuming that the events happen randomly and independently (Laguna & Marklund, 2019, p. 202).

For each team, the number of expected goals λ (home team) and μ (away team) are calculated using the given offensive α and defensive β parameters. A home effect γ and δ is added if Qatar is amongst the teams playing, and set to "False" if not. The values of expected goals for home team λ and away team μ are then used to calculate the probability of each team scoring X_{ij} and Y_{ij} goals, between 0 and 5 goals. As the task restricts the scorelines between 0 and 5 goals (removing possible scores above this), we then have to normalize the probabilities to ensure they sum up to 1.

To simulate a match, a score is randomly drawn for each team based on the pre-calculated probabilities. This means the more probable scores will occur more often, and ensures the simulation reflects realistic outcomes. After all matches in a group are simulated, the points are awarded to each team, and the table is updated accordingly. The table is sorted based on the rules of the competition, such as points and goal difference (Wikipedia contributors, 2026)

This process is repeated for 10 000 iterations, with results added to a dictionary. These results are then divided by iterations to get the probability of occurrence.

1.2 Optimalization

Before simulation, I added a function to compute the probabilities for each team's score before the Monte Carlo simulation. This was done to avoid calculation being done in every iteration, as my program used 8 minutes to run 10 000 calculations without doing this.

1.3 Results

The results of the Monte Carlo simulation are shown in Figure 1. The figure shows the probabilities of each team winning and qualifying in their group, as well as mean expected points and goal difference through the group stage. We can see that some groups are evenly distributed regarding qualifying, with similar probabilities for each team. Other groups, such as group E have clear favorites. The host nation, Qatar, has the second lowest probability of qualifying, only beaten by Cameroon, even with home effect applied.

1	
2	
3	Group A
4	
5	NET: {'First place probability': 0.6971, 'Qualifying probability': 0.9153, 'Expected points': 6.8531, 'Expected goal difference': 4.3199}
6	SEN: {'First place probability': 0.0796, 'Qualifying probability': 0.3295, 'Expected points': 3.3445, 'Expected goal difference': -1.1018}
7	ECU: {'First place probability': 0.2014, 'Qualifying probability': 0.6415, 'Expected points': 4.6909, 'Expected goal difference': 0.9528}
8	QAT: {'First place probability': 0.0219, 'Qualifying probability': 0.1137, 'Expected points': 1.8581, 'Expected goal difference': -4.1709}
9	
10	
11	Group B
12	
13	ENG: {'First place probability': 0.6374, 'Qualifying probability': 0.8666, 'Expected points': 6.2352, 'Expected goal difference': 2.8361}
14	USA: {'First place probability': 0.1501, 'Qualifying probability': 0.4335, 'Expected points': 3.651, 'Expected goal difference': -0.6169}
15	IRN: {'First place probability': 0.1345, 'Qualifying probability': 0.4265, 'Expected points': 3.5938, 'Expected goal difference': -0.6013}
16	WAL: {'First place probability': 0.078, 'Qualifying probability': 0.2734, 'Expected points': 2.8251, 'Expected goal difference': -1.6179}
17	
18	
19	Group C
20	
21	ARG: {'First place probability': 0.6861, 'Qualifying probability': 0.9076, 'Expected points': 6.7557, 'Expected goal difference': 3.9119}
22	KSA: {'First place probability': 0.0165, 'Qualifying probability': 0.0994, 'Expected points': 1.6795, 'Expected goal difference': -3.7717}
23	MEX: {'First place probability': 0.1481, 'Qualifying probability': 0.5024, 'Expected points': 4.1237, 'Expected goal difference': -0.0549}
24	POL: {'First place probability': 0.1493, 'Qualifying probability': 0.4906, 'Expected points': 4.1021, 'Expected goal difference': -0.0853}
25	
26	
27	Group D
28	
29	FRA: {'First place probability': 0.6652, 'Qualifying probability': 0.9106, 'Expected points': 6.7101, 'Expected goal difference': 3.9171}
30	AUS: {'First place probability': 0.0486, 'Qualifying probability': 0.222, 'Expected points': 2.6719, 'Expected goal difference': -2.3393}
31	DEN: {'First place probability': 0.2488, 'Qualifying probability': 0.69, 'Expected points': 4.9227, 'Expected goal difference': 1.1188}
32	TUN: {'First place probability': 0.0374, 'Qualifying probability': 0.1774, 'Expected points': 2.3763, 'Expected goal difference': -2.6966}
33	
34	
35	Group E
36	
37	ESP: {'First place probability': 0.5848, 'Qualifying probability': 0.8734, 'Expected points': 6.4374, 'Expected goal difference': 3.5234}
38	CRC: {'First place probability': 0.0184, 'Qualifying probability': 0.0926, 'Expected points': 1.7099, 'Expected goal difference': -3.8959}
39	GER: {'First place probability': 0.3097, 'Qualifying probability': 0.7199, 'Expected points': 5.2522, 'Expected goal difference': 1.6388}
40	JAP: {'First place probability': 0.0871, 'Qualifying probability': 0.3141, 'Expected points': 3.3663, 'Expected goal difference': -1.2663}
41	
42	
43	Group F
44	
45	BEL: {'First place probability': 0.5551, 'Qualifying probability': 0.8285, 'Expected points': 6.1074, 'Expected goal difference': 3.0045}
46	CAN: {'First place probability': 0.013, 'Qualifying probability': 0.0707, 'Expected points': 1.4303, 'Expected goal difference': -4.1598}
47	MAR: {'First place probability': 0.2354, 'Qualifying probability': 0.5915, 'Expected points': 4.6887, 'Expected goal difference': 0.8056}
48	CRO: {'First place probability': 0.1965, 'Qualifying probability': 0.5093, 'Expected points': 4.3608, 'Expected goal difference': 0.3497}
49	
50	
51	Group G
52	
53	BRA: {'First place probability': 0.7919, 'Qualifying probability': 0.9562, 'Expected points': 7.2938, 'Expected goal difference': 4.6726}
54	SRB: {'First place probability': 0.0693, 'Qualifying probability': 0.3865, 'Expected points': 3.5396, 'Expected goal difference': -0.9236}
55	SUE: {'First place probability': 0.1313, 'Qualifying probability': 0.5809, 'Expected points': 4.3552, 'Expected goal difference': 0.3017}
56	CMR: {'First place probability': 0.0075, 'Qualifying probability': 0.0764, 'Expected points': 1.4964, 'Expected goal difference': -4.0507}
57	
58	
59	Group H
60	
61	POR: {'First place probability': 0.5852, 'Qualifying probability': 0.8573, 'Expected points': 6.261, 'Expected goal difference': 3.1287}
62	GHA: {'First place probability': 0.0387, 'Qualifying probability': 0.1631, 'Expected points': 2.2233, 'Expected goal difference': -2.8339}
63	URU: {'First place probability': 0.2832, 'Qualifying probability': 0.6753, 'Expected points': 4.9079, 'Expected goal difference': 1.1563}
64	KOR: {'First place probability': 0.0929, 'Qualifying probability': 0.3043, 'Expected points': 3.1946, 'Expected goal difference': -1.4511}

Figure 1: Simulation results

1.4 Model evaluation

The model gives a reasonable approach to estimating probabilities for the outcomes of the group stage. The Poisson distribution is fitting as it models the unpredictable nature of football, with reoccurring events (goals) happening in a given interval. Variation in team strength is taken into account with the offensive and defensive parameters α and β , as is playing at home for Qatar with parameters γ and δ . Simulating over 10 000 iterations gives a good estimate of the probabilities, as it reduces the influence of outliers and fluctuations in the results, together with the goal cap set at 5.

However, the model is a simplification of reality. It assumes that goals are independent events, failing to take into account the many factors that can influence the outcome of a single match. Additionally, the parameters are set to be constant, even

though factors such as injuries, tactical decisions and even the psychological effect of having supporters may have an impact on this, and therefore also on the results. While the model provides useful estimates of probabilities, it should be viewed as an approximation rather than an exact prediction of the group stage.

2 Task 2: The business challenge

2.1 AS-IS process architecture

2.1.1 Defining boundaries

The Freight Claim Request (FCR) is first triggered when a client experiences a cargo loss and files a formal claim. The process officially starts when the client submits the FCR at the GCI's central intake desk. The process concludes either when the claim is declined due to incomplete documentation or suspected fraud, or when The Treasury Department releases the fund backs to the client.

2.1.2 Flow unit

The main flow unit moving through this process is the Freight Claim Request (FCR) file. This includes the damage reports, photographic evidence, original shipping manifests, and financial valuation documents.

Resources	Buffers
Administrative clerk (at least 1)	The central intake desk
ClaimFlow IT system	Digital queue, marine investigation department
Marine investigators (currently 3)	Finance department queue
Senior claims adjusters (currently 2)	Legal queue
Lawyer (currently 1)	Treasury department queue
Treasury department	Clerk's inbox delay

Table 1: Resources and buffers in the claims process

For a full visual mapping of where these resources and buffers operate across the customer/front-office/back-office domains, please see the General Process Chart and Service System Map (SSM) in appendix A.

2.1.3 Classifications

To evaluate the activities in the AS-IS process we have classified each in value-adding, non-value-adding and business value-adding. As we can see in the table under only the fund transfer directly creates value for the customer, as it is delivering the financial compensation that is expected from the claim. Most of the other activities are classified as BVA because they are necessary for meeting regulatory requirements. While several activities marked by NVA because they represent delays, transportations or rework, which does not create customer value.

Activity	Type	Reasoning
Intake and Requisite Review	BVA	Administrative validation claim
Digital transfer	NVA	Pure system transport (delay)
Damage Investigation Study	BVA	It is required to determine liability
Liability decision	BVA	Insurance control decision
Handoff to finance	NVA	Transfer between departments
Financial assessment	BVA	Calculating valuations
Return transfer	NVA	Rework transport
Clerk inbox	NVA	Waiting time
Correction	NVA	Rework caused by errors
Re-submission	NVA	Administrative transfer
Payout approval	BVA	Fraud control
Settlement communications	BVA	Legal requirement
Handoff to treasury	NVA	Transfer between departments
Funds transferred	VA	Customer receives payments

Table 2: Classification of activities in the AS-IS claims process

2.1.4 Case for action

The current AS-IS process suffers from systemic inefficiencies and functional silos driven by a high volume of NAV activities. Multiple handoffs between departments introduce significant delays and unproductive use of time into the workflow. Most critically the 25

2.1.5 Vision statement

The redesigned process will help eliminate departmental barriers by adopting a cross-functional case management approach and implement mistake proofing at the initial intake. Being able to capture the correct currency data at the source, and streamlining automated handoffs, the new design will help drastically reduce the total cycle time and ensure a more efficient payout experience for the client.

2.1.6 KPI

Average cycle time

Measuring the average time from when a FCR is submitted, until the claim is either paid out or rejected. It reflects the overall cycle time of the process and identifies where delays are caused by rework, queues or inefficient handoffs. The goal is to reduce the total cycle time to address customer complaints about delays.

2.1.7 Quality

This KPI is measuring the percentage of claims that go through the whole process without entering the rework loop. It reflects the reliability of the process from the customers perspective and is an indicator of how consistently GCI deliver timely claim settlements.

2.1.8 Cost: Resource utilization

This KPI measures the percentage of time senior claim adjusters are actively working on a claim. The goal is to ensure that expensive resources are used effectively and not wasted on administrative delays or departmental handoffs.

2.2 Workflow documentation

To ensure an accurate simulation of the AS-IS process, we have organized all the points from the case description and their respective process steps. This data will be used for the logic in the JaamSim model, the flowchart diagram and table of documented data can be found in appendix B.

2.3 Baseline analytic analysis

To be able to identify the bottlenecks we will compare demand with capacity for each resource. We assume an 8-hour workday, which is 480 minutes. The arrival rate λ is 1 claim each 120 minutes, which is 4 claims in a day. The calculation for demand is shown in the excel file. Let's take legal as an example:

$$\text{Legal: } 4 \times 0.90 \times 0.95 = 3.249 \text{ claims}$$

Resource	number of resource (c)	Time per unit (T)	Capacity/day ($480/T \cdot C$)	Demand
Intake clerk	1	20	24	4
Marine invest.	3	1440	1	3,80
Senior adjust.	2	480	2	4,275
Legal (lawyer)	1	480	1	3,249

Figure 2:

The most critical findings we did is that three out of the four groups of resources are operating with utilization rate exceeding 100%. The primary bottleneck is the marine investigators at 380% utilization, while legal and senior adjusters also operate well over capacity. On the other hand, the intake clerk is severely under-utilized. This over utilization is worsened by high variability in both claim arrivals and processing times, creating massive and unrecoverable backlogs. Furthermore, functional silos continuous friction at handoff, and a 25% rework loop adds over three days of NVA delay to affected claims. At last, these mistakes create structural flaws a “death spiral” of continuous increasing cycle times, inflated cost and customer dissatisfaction.

2.3.1 TO - BE process design

The conceptual redesign of the FCR transitions GCI from a fragmented functional orientation to a product-oriented workflow. By focusing on the claim rather than the departmental boundaries, the new design achieves significant improvements in quality and cycle time. You can see the redesigned process diagram in appendix C. The following principles have been applied:

2.3.2 Organize work around outcomes, not tasks

We dismantled functional silos by introducing a cross-functional management approach. Instead of performing isolated departmental tasks, resources coordinated by a case manager are organized into claim resolution teams that take shared responsibility for the ultimate outcome, this being a quick and accurate payout.

2.3.3 Capture information at the source

We implemented an automated system validation at the initial intake to ensure that the correct currency exchange rate is captured immediately. This will help guarantee data quality upfront adding to the principle of capturing information once, and eliminate the need for manual re-entry later.

2.3.4 Design for dominant flow, not expectations and mistake proofing critical steps

Using the strategy “Poka-Yoke”, which is a “well-known strategy for and improvement in product and process design” (Laguna & Marklund, 2019, p. 99), at the intake step completely exterminates the 25% currency rework loop, and its 3 day inbox delay. The workflow is now optimized for the dominant flow, which is 90% of the valid claims. This is removing complex exception routes and unnecessary work-in-progress buffers.

2.4 Simulation analysis: AS-IS vs TO-BE

2.4.1 Structure

The AS-IS model represents the current state of GCI’s claim handling. This is characterized by significant time buffers and sequential processing. The model follows a sequential structure where each task must be fully completed before the next department can begin.

The main simulation entity is the FCR. Each claim represents one claim submitted by a client and moves through the process until its either rejected or paid out. Each entity contains a limited number of attributes used to track the progress through the system. The model includes the main processing activities described in the case: Intake and requisite review, digital transfer, damage investigation, financial assessment, rework loop, settlement communication and funds transfer. While state variables include queue length, utilization and number of entities completed or rejected.

The AS-IS model process is severely hindered by some structural flaws. Firstly, the strictly sequential handoffs between isolated departments introduce significant non-value-adding delays. Second, a 25% rework loop forces claims back to intake, creating a massive bottleneck that inflates total cycle time. Finally, we see that the process suffers from rigid resource dependencies, because tasks rely on a single, specialized resources working in functional silos, creating queueing and work in progress accumulation.

2.4.2 Arrivals and service times

The arrival process is modeled using an exponential distribution with a mean of 120 minutes. However, to be able to catch the true process dynamics, service times are modeled using the specific stochastic distributions provided, rather than simple averages.

2.4.3 Validation checks

The model logic was visually verified in JaamSim to ensure that the routing behavior at the Branch objects, matches the conceptual flow chart. The utilization state variable for marine investigators, causing the queue length s to grow continuously due to capacity constraints. Consequently, the average cycle time continuously inflates, proving that the current system is mathematically unstable. The simulation's accuracy was also verified using Little's Law.

$$L = \lambda \times W$$

By comparing the recorded average queue length against the calculated arrival rate and observed wait times at the bottleneck, we see that the model demonstrates a high degree of internal validity and mathematical consistency.

$$L \approx \left(\frac{\text{Number added}}{\text{SimTime}} \right) \times \text{AverageQueueTime}$$

Where L is Queue Length Average

Example at 550 hours:

$$81.06 \approx \left(\frac{241}{550.25} \right) \times 185.077$$

$$81.06 \approx 81.06$$

2.4.4 TO-BE, improvements

The redesigned model transitions from a sequential workflow to one utilizing parallel processing and case management. After the initial intake, a duplicate object splits the FCR, sending it to marine investigation and financial assessment simultaneously. Later a combine object synchronizes the two parallel paths before the claim proceeds to the legal settlement stage.

2.4.5 Key assumptions

To initiate mistake proofing an automated validated step is assumed at the initial digital intake. This «Poke-yoke» mechanism correctly captures correct currency data at the source. This is eliminating the 25% rework branch, and the associated multi-day delays.

By utilizing a cross functional case Management approach, the model assumes that the 15-minute departmental handoffs are reduced to zero, helping remove non-value-adding friction. As a recommendation, to prevent the bottleneck from shifting

downstream, we chose to increase the capacities for the lawyer and investigators (investigators, $n=5$, lawyer, $n=2$).

2.4.6 Comparison

The simulation results show that by implementing additional investigators and parallelizing the workflow, the average waiting time in the investigation queue dropped from 185h to 127h. This reduction proves that the redesigned process is more resilient to variability in arrival rates. Even though the waiting time decreased, there is still a residual queue due to the high arrival rates, and it shows the need for continuous monitoring of capacity. We also see that the average units in use of all investigators, lawyers and adjuster have increased in the new model, resulting in higher utilization. Further we see that the number of processed claims in the same amount of time in the TO-BE simulation almost doubled from 57 claims to 105 claims.

Simulation shows that the redesigned process significantly improves the system. Ultimately this structural redesign reduces the average cycle time, transforming GCI's claims handling into a more efficient, scalable and customer focused operation.

Appendix

Appendix A

Activity	Count	Time (min)	% of total time
Operation	5	3390	85,50 %
Inspection	1	20	0,50 %
Transportation	5	75	1,89 %
Delay	1	480	12,11 %
Storage	0	0	0,00 %
Total	12	3965	100,00 %

Figure 3: General process chart

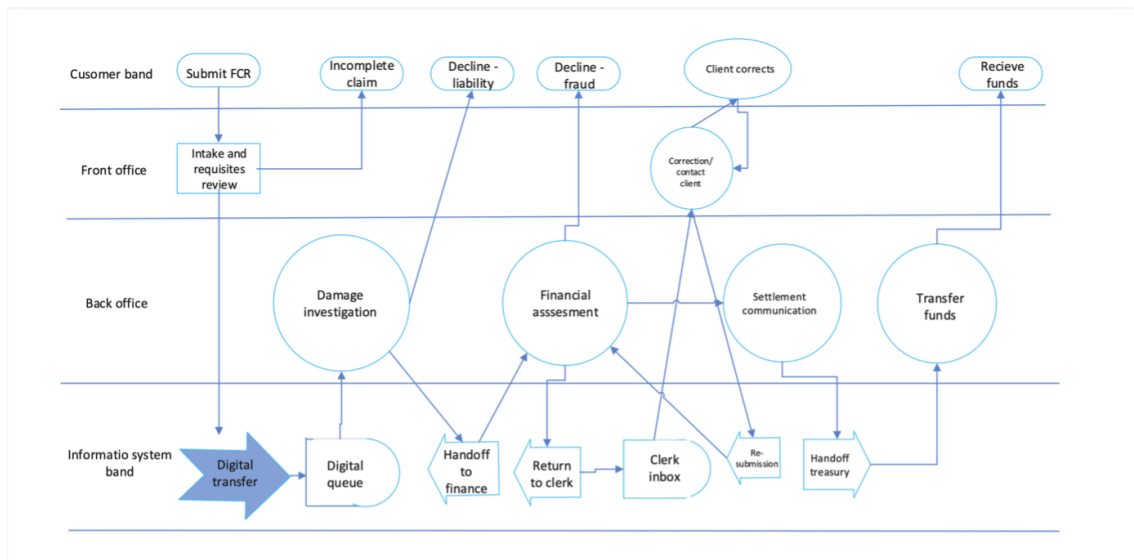


Figure 4: Service System Map (SSM)

Appendix B

Flowchart ID	Process activity	Resource	Statistical distribution/Data	Simulation usage
0	Arrival of claim	–	Exponential (mean = 120 min)	Inter-arrival time for Entity-Generator
1	Intake review	1 clerk	Normal ($\sigma = 4$, $\mu = 20$) min	Service time for Server object
1a	Intake decision	–	5% reject / 95% proceed	Branch logic and choice probabilities
2	Digital transfer	IT system	Gamma ($k = 5$, $\theta = 3$), mean 15 min	Processing time for delay object
3	Damage investigation	3 investigators	Gamma ($\alpha = 2$, $\beta = 720$), mean 1440 min	Service time for server
4	Liability decision	–	10% reject / 90% OK	Branch logic and choice probabilities
5	Handoff to finance	–	Constant 15 min	Entity delay between stations
6	Financial assessment	2 adjusters	Beta ($\alpha = 2$, $\beta = 2$), 360–600 min	Service time
7a	Rework check	–	25% rework / 75% OK	Loop back and branch logic
7b	Return transfer	IT system	Constant 15 min	Return transfer delay
7c	Clerk inbox delay	Clerk	Erlang ($k = 3$, $\theta = 160$), mean 480 min	Delay before correction
7d	Correction	Clerk/client	Normal ($\mu = 960$, $\sigma = 120$) min	Service time for rework station
7e	Re-submission	IT system	Constant 15 min	Re-submission delay
8	Payout approval	–	5% decline / 95% approved	Branch object
9	Settlement draft	Lawyer	Uniform (420–540) min	Service time for final bottleneck
10	Handoff to treasury	–	Constant 15 min	Treasury handoff delay
11	Funds transfer	Treasury	Normal ($\mu = 30$, $\sigma = 5$) min	Final value-adding process

Table 3: Detailed workflow documentation

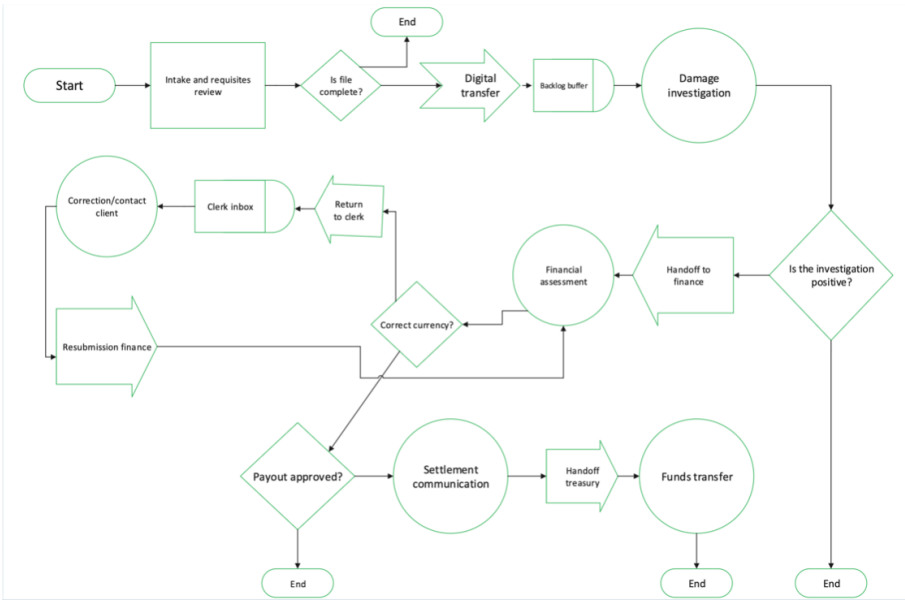


Figure 5: Detailed flowchart

Appendix C

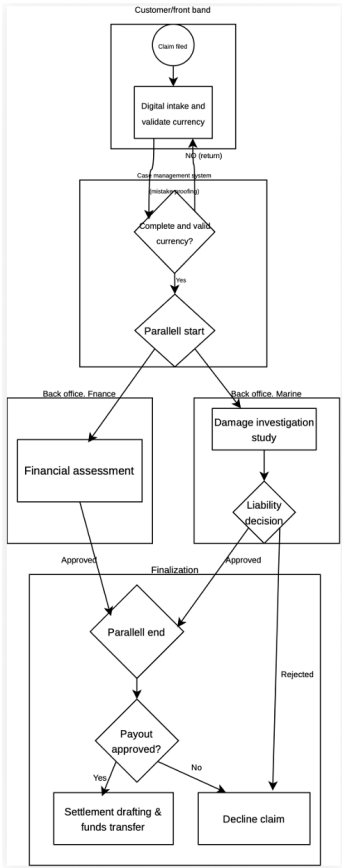


Figure 6: Appendix C

References

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